3 17 18 19 20 21 27 28 30 44 50 64 68

第二章

2.3 $\begin{cases} T-\mu_1 m_A g = m_A a \\ F-T-\mu_1 m_A g-\mu_2(m_B+m_A)g = m_B a \end{cases}$

$\Rightarrow F=(m_A+m_B)a+2\mu_1 m_A g+\mu_2(m_A+m_B)g$

$\Rightarrow F=13.2N$

2.12 $\begin{cases} T_2\cos\theta = mg \\ T_1\cos\theta = T_2\cos\theta+mg \\ T_1\sin\theta+T_2\sin\theta = m\omega^2 l\sin\theta \end{cases}$

$\Rightarrow \begin{cases} T_1 = \frac{2}{3}m\omega^2 l \\ T_2 = \frac{1}{3}m\omega^2 l \\ \theta = \arccos\frac{3g}{\omega^2 l} \end{cases}$

2.18 $T-(T+dT) = \frac{m dr}{l}\omega^2 r$

$\Rightarrow -dT = \frac{m\omega^2}{l} r\,dr$

$\Rightarrow \int_T^0 -dT = \int_r^l \frac{m\omega^2 r}{l}dr$

$\Rightarrow T = \frac{m\omega^2(l^2-r^2)}{2l}$

2.19 (1) $\begin{cases} T\sin\theta = m\omega^2 l\sin\theta \\ T\cos\theta = mg+ma_0 \end{cases} \Rightarrow \begin{cases} T = \frac{mg+ma_0}{\cos\theta} \\ \omega = \sqrt{\frac{g+a_0}{l\cos\theta}} \end{cases}$

$\Rightarrow \begin{cases} T = 7.22N \\ \omega = 7.6\,rad/s \end{cases}$

2.20 $\begin{cases} (Mg+F_N)\sin\alpha = Ma_A \\ (mg-F_N) = ma_B \\ a_A\sin\alpha = a_B \end{cases} \Rightarrow \begin{cases} a_A = \frac{Mg+mg}{M+m\sin^2\alpha}\sin\alpha \\ a_B = \frac{Mg+mg}{M+m\sin^2\alpha}\sin^2\alpha \\ F_N = \frac{Mmg\cos^2\alpha}{M+m\sin^2\alpha} \end{cases}$

A对地面加速度大小 $a_A = \frac{Mg+mg}{M+m\sin^2\alpha}\sin\alpha$，方向与水平呈$\alpha$角向下

B对A加速度大小 $a_{BA} = a_A\cos\alpha = \frac{Mg+mg}{M+m\sin^2\alpha}\sin\alpha\cos\alpha$，方向水平向右

2.21 $\begin{cases} m_1 g-T-m_1 a_0 = m_1 a \\ T+m_2 a_0-m_2 g = m_2 a \end{cases} \Rightarrow \begin{cases} a = \frac{m_1 g+m_2 a_0-m_2 g-m_1 a_0}{m_1+m_2} \\ T = \frac{m_1 m(g-a_0)}{m_1+m_2} \end{cases}$

2.27

$$\begin{cases} f\sin\alpha + mg = F_N\cos\alpha \\ f\cos\alpha + F_N\sin\alpha = m\omega^2 l_1\cos\alpha \\ f = \mu F_N \end{cases}$$

$$\Rightarrow l_1 = \frac{(\mu\cos\alpha+\sin\alpha)g}{(\cos\alpha-\mu\sin\alpha)\omega^2\cos\alpha}$$

$$\begin{cases} mg = F_N\cos\alpha + f\sin\alpha \\ F_N\sin\alpha - f\cos\alpha = m\omega^2 l_2\cos\alpha \\ f = \mu F_N \end{cases}$$

$$\Rightarrow l_2 = \frac{(\sin\alpha-\mu\cos\alpha)g}{(\cos\alpha+\mu\sin\alpha)\omega^2\cos\alpha}$$

$$l_2 \le l \le l_1$$

2.28 1° $m_2g - m_1g - m_2a \le 0$ 2° $m_2g - m_1g - m_2a > 0$

$$\begin{cases} f = T \\ m_2g - f = m_2(a - a_0) \\ m_1g - T = m_1a_0 \end{cases} \qquad \begin{cases} f = T \\ m_2g - f = m_2(a + a_0) \\ T - m_1g = m_1a_0 \end{cases}$$

$$\Rightarrow \begin{cases} a_{物} = a_0 = \frac{m_2a + m_2g - m_1g}{m_1+m_2}, \text{方向向下} \\ a_{相} = a - a_0 = \frac{m_1a - m_1g + m_2g}{m_1+m_2} \text{方向向下} \\ T = \frac{2m_1m_2g - m_1m_2a}{m_1+m_2} \end{cases} \qquad \begin{cases} a_{物} = a_0 = \frac{m_2g - m_1g - m_2a}{m_1+m_2}, \text{方向向上} \\ a_{相} = a + a_0 = \frac{m_2g - m_1g + m_1a}{m_1+m_2} \text{方向向下} \\ T = \frac{2m_1m_2g - m_1m_2a}{m_1+m_2} \end{cases}$$

2.30 (1) $I = \int_{t_0}^{t_1} F\,dt \Rightarrow I = 6\,N\cdot s$

$$\bar{F} = \frac{I}{\Delta t} \Rightarrow \bar{F} = 15\,N$$

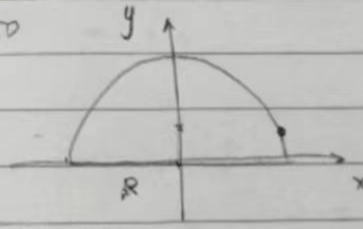
(2) $I = mv_1 - mv_0 \Rightarrow v_1 = 3\,m/s$

2.44

$$\begin{cases} t = \frac{l\sin\alpha}{g} \\ (m+M)v_0\cos\alpha = m(v-N) + Mv \\ x = vt \end{cases} \Rightarrow x = \frac{[(m+M)v_0\cos\alpha + mv]l\sin\alpha}{(m+M)g}$$

$$\begin{cases} t = \frac{l\sin\alpha}{g} \\ (m+M)v_0\cos\alpha = m(v-N) + Mv \\ x = vt \end{cases} \Rightarrow x = \frac{[(m+M)v_0\cos\alpha - mv]l\sin\alpha}{(m+M)g}$$

2.50



$$y = \frac{1}{\frac{1}{2}\pi R^2}\int_0^R 2y\sqrt{R^2-y^2}\,dy$$

$$= \frac{2}{\pi R^2}\cdot\frac{2}{3}R^3$$

$$= \frac{4R}{3\pi}$$

即 $(0, \frac{4R}{3\pi})$ 处

2.64 $F\Delta t = (m_1+m)V_2 - m_1V_1$

$$\Rightarrow F = \frac{(m_1+m)V_2 - m_1V_1}{\Delta t} \Rightarrow F = 1.08\times10^{4}\,N$$

2.68 $\mu dt V_b + mV_a = (m+\mu dt)(V_a+dv)$

$$\Rightarrow a = \frac{dv}{dt} = \frac{\mu(V_b - V_a)}{m}$$

2.68